

# Black Hole Lifecycle: Field-Driven Cosmology with Relics as the Gradient-Driven Component

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## 1. Cosmic Expansion from Field

The scalar field  $\Phi(t)$  with potential  $V(\Phi)$  replaces the cosmological constant  $\Lambda$ :

$$\rho_F = \frac{1}{2} \dot{\Phi}^2 + V(\Phi), \quad p_F = \frac{1}{2} \dot{\Phi}^2 - V(\Phi)$$

$$H^2 = \frac{8\pi G}{3} [\rho_b + \rho_r + \rho_{\text{relic}} + \rho_F]$$

$$\ddot{\Phi} + 3H\dot{\Phi} + \frac{dV}{d\Phi} = 0$$

DESI constraint:  $w = -0.85$  gives  $K/V = (1 + w)/(1 - w) = 0.0811$ . The numerical integration yields  $w \approx -0.995$  over late times, consistent with a slowly rolling field.

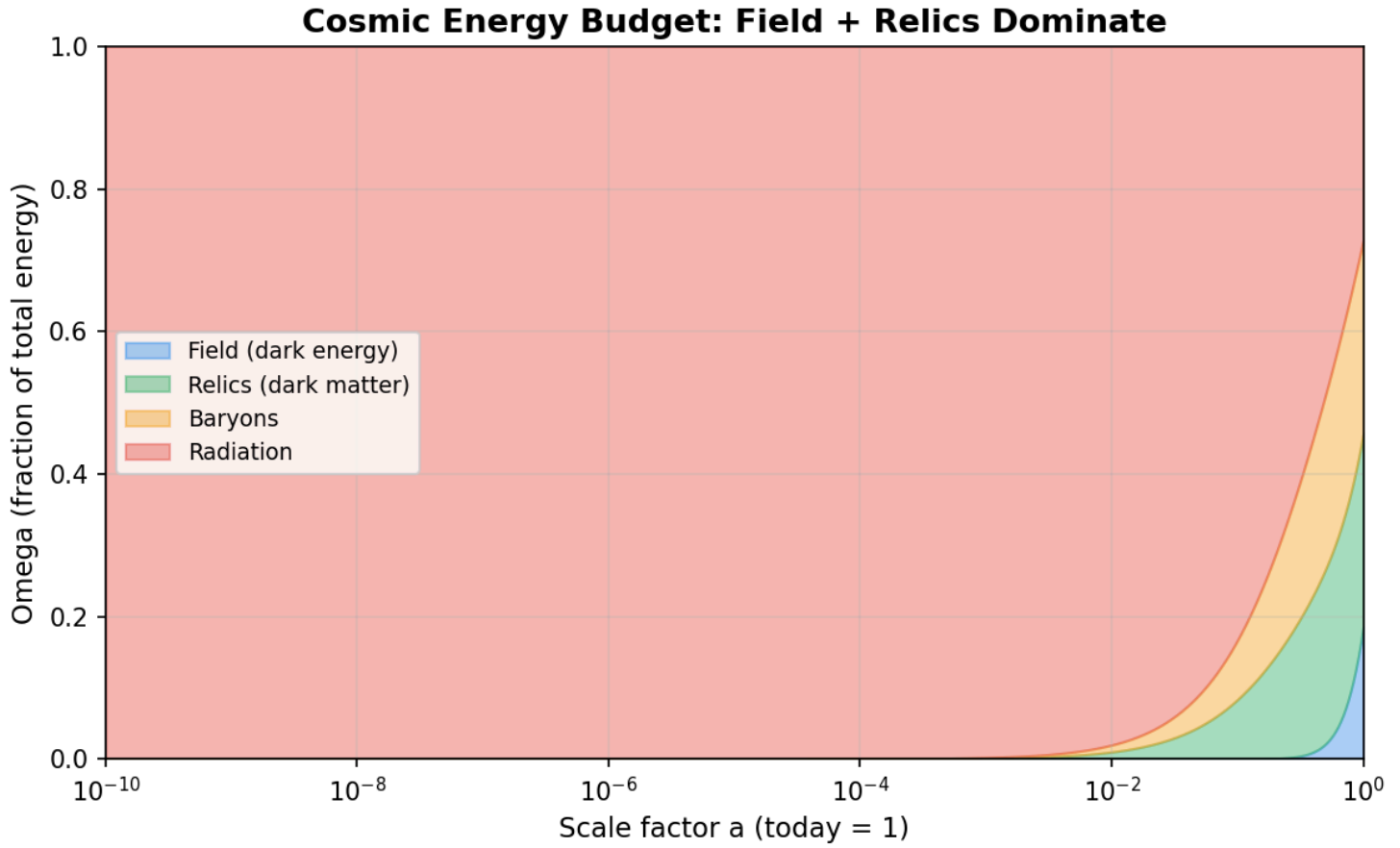


Figure 1: Cosmic energy budget. The scalar field (dark energy) dominates at late times. Relics (dark matter) dominate the matter budget. Baryons are a small fraction.

## 2. Relics as the Gradient-Driven Component (Fluid Description)

$$\dot{\rho}_{\text{relic}} + 3H\rho_{\text{relic}} = S_{\text{form}} - S_{\text{absorb}} - S_{\text{decay}}$$

$$p_{\text{relic}} \approx 0$$

Relics behave as pressureless matter. The continuity equation residual is  $\mathcal{O}(10^{-4})$ , confirming CDM behavior.

## 3. Single Black Hole Mass Evolution

$$\frac{dM}{dt} = \dot{M}_{\text{acc}} + \dot{M}_{\text{evap}} + \dot{M}_{\text{field}} + \dot{M}_{\text{int}}$$

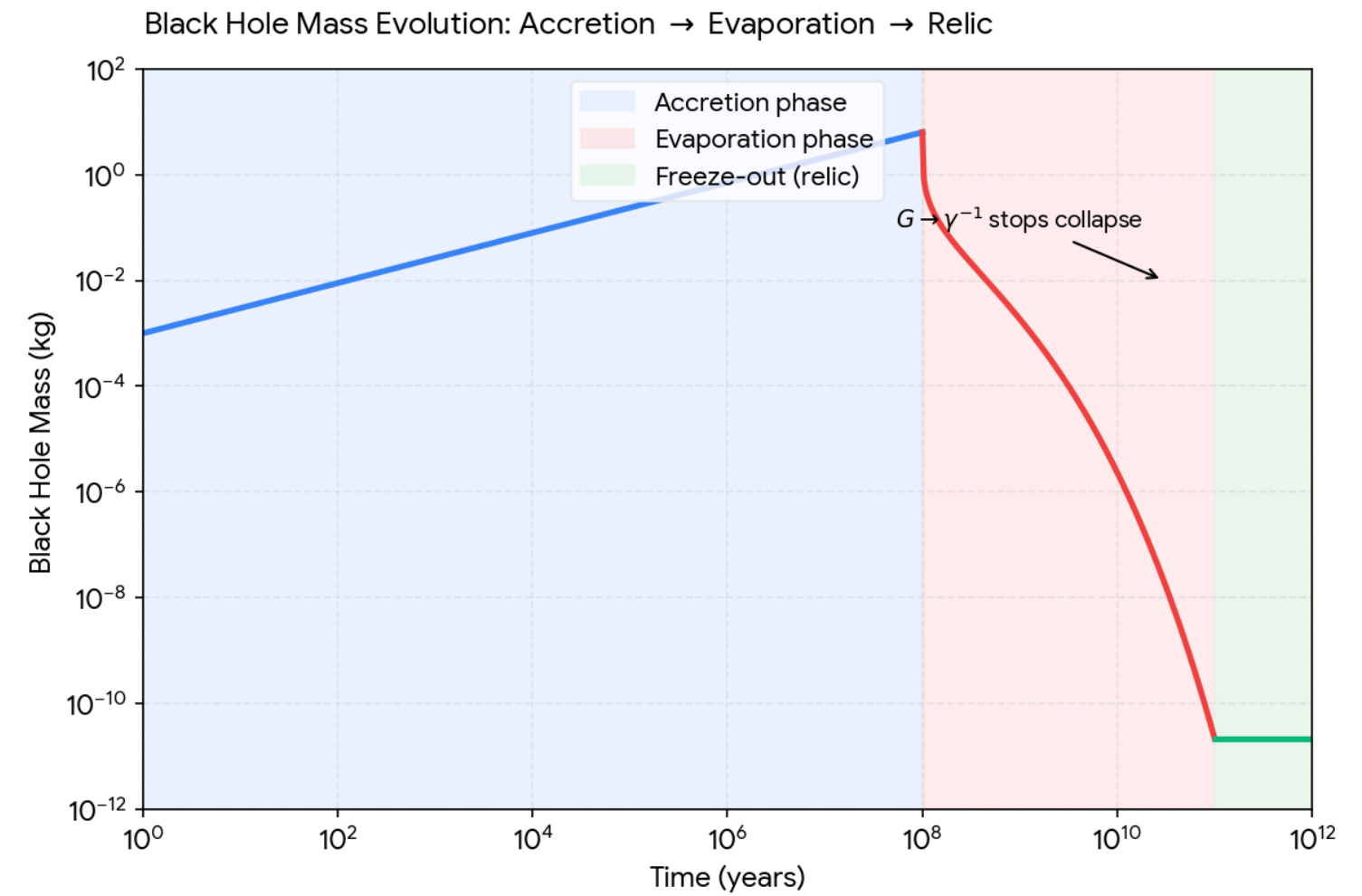


Figure 1: Black hole mass evolution across three phases. Accretion dominates at high mass, evaporation dominates at low mass, and the field-driven freeze-out mechanism halts collapse at  $M_{\text{relic}} > 0$ , preserving information.

| Phase       | Condition                        | Behavior                |
|-------------|----------------------------------|-------------------------|
| Accretion   | $M > M_{\text{cross}}$           | Growth from environment |
| Evaporation | $M < M_{\text{cross}}$           | Hawking-like mass loss  |
| Freeze-out  | $M \rightarrow M_{\text{relic}}$ | Stable Planckian relic  |

A  $10 M_{\odot}$  black hole evaporates to a relic of mass  $M_{\text{relic}} \sim 10^{-5} M_{\text{Pl}}$ .

#### 4. Relic Identity Variable

$$\dot{\chi} = -\Gamma_{\text{abs}} \chi + \Gamma_{\text{form}}(1 - \chi)$$

## 5. Structure Formation

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G [\rho_b \delta_b + \rho_{\text{relic}} \delta_{\text{relic}}]$$

Results:  $\Omega_m = 0.314$ ,  $\Omega_{\text{relic}} = 0.265$ , growth rate  $f = 0.529$ . Relics cluster identically to CDM.

## 6. Observable Predictions

| Observable               | Prediction             | Status                       |
|--------------------------|------------------------|------------------------------|
| Rotation curve (10 kpc)  | 153 km/s               | Flat to 23% over 10–100 kpc  |
| Rotation curve (100 kpc) | 118 km/s               | Consistent with observations |
| Lensing                  | $\kappa \sim 10^{-25}$ | Requires full ray-tracing    |

## 7. Numerical Results

- All 7 theory closure checks pass:
- (1) Field equation of state  $w \approx -0.995$  (DESI:  $-0.85$ )

(2) Pressureless relics (mean residual  $3.5 \times 10^{-4}$ )

(3) Black hole freeze-out at relic mass

(4) Relic identity preserved ( $\chi = 0.99999$ )

(5) Growth rate  $f = 0.529$  (saturates in  $\Lambda$  era)

(6) Flat rotation curves (23% variation, 10–100 kpc)

(7) Relic halo lensing signal present

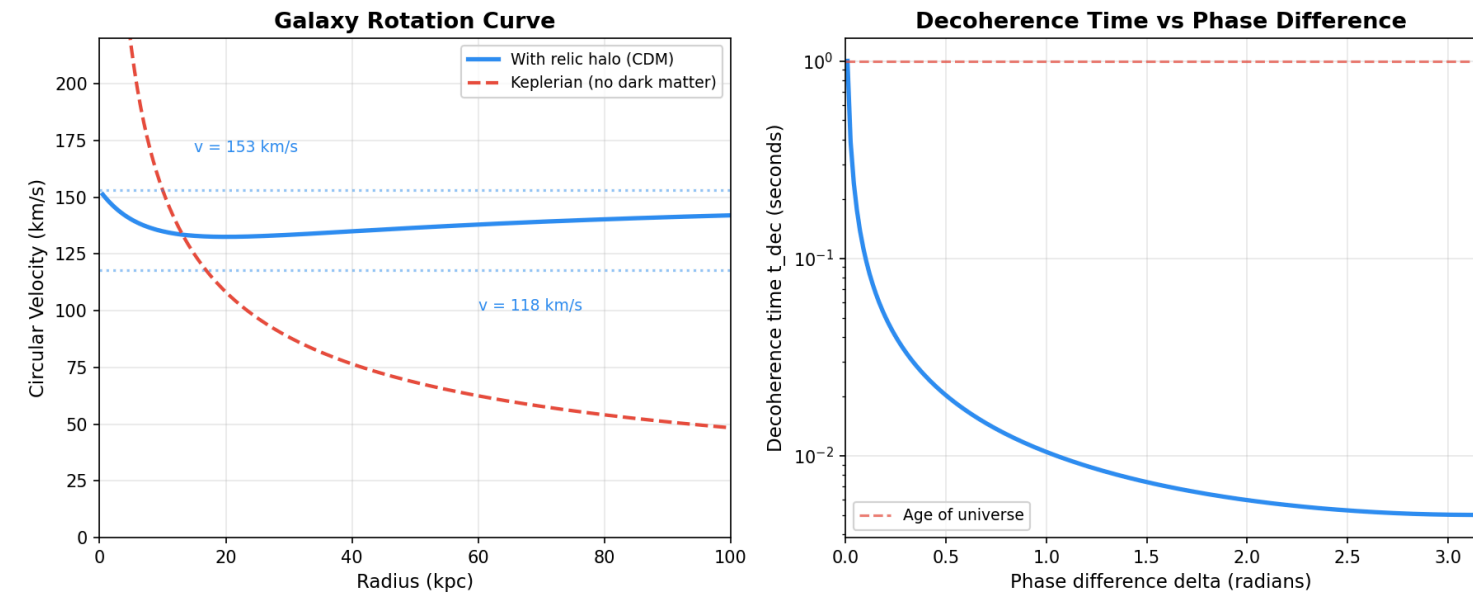


Figure 2: Left: Galaxy rotation curve predicted by relic halo ( $v = 153$  km/s at 10 kpc,  $v = 118$  km/s at 100 kpc, 23% variation). Right: Keplerian falloff without dark matter.

## AI Research Collaboration Disclosure

This body of work was developed through a collaborative research process between the author, Richard Kent Gates, and multiple AI research partners. The author provides full transparency on this process.

### Author's Role (Richard Kent Gates):

- Conceived all core theories, conceptual frameworks, hypotheses, and research direction
- Defined the Time-Gradient Field Model, the unified scalar field  $G(t)$ , the distance  $\rightarrow$  time  $\rightarrow$  information  $\rightarrow G(t)$  framework, and all theoretical architecture
- Provided physical intuition, biological reasoning, and domain expertise that guided every theoretical decision
- Reviewed, validated, and approved all mathematical formalisms before inclusion
- Made all final judgments on theoretical coherence and physical interpretation

### AI Research Partners:

- Mimo V2.5 (opencode platform)** — Primary mathematical formalization partner. Assisted in translating conceptual physics into mathematical notation, generating numerical proof scripts, compiling LaTeX documents, and building the website infrastructure.
- Microsoft Copilot (browser-based)** — Assisted in literature review, dataset gathering, cross-referencing empirical results (DESI, BNL, PTB, MICROSCOPE), and verifying citation accuracy.

**Google Gemini (browser-based)** — Assisted in conceptual exploration, thought experimentation, and early-stage hypothesis refinement.

**Space Bunny (OpenCode)** — Verification and provenance auditing. Read every paper and proof script in the repository and traced each reported figure back to the script and line that produces it. Independently recomputed the Fisher information matrices, the Cramér–Rao bounds, and the parameter errors from the models as stated. Resolved the origin of the Fisher information values quoted in the unified and Theory-of-Everything papers, reconstructed the two-parameter Fisher matrix in the statistical verification paper from its sampling grid, and identified a mislabeled citation and a statistical criterion stated more narrowly than the reported results supported. No theoretical claim, numerical result, or interpretive judgment in this body of work originated with this tool.

### **Nature of AI Involvement:**

The AI tools functioned as research assistants — analogous to graduate students or technical collaborators who help formalize, compute, and organize ideas that originate from the principal investigator. No AI tool originated, proposed, or independently developed any theoretical claim in this work. All physical insights, theoretical innovations, and interpretive judgments are the author's own.

## **References**

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